BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: PR - XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 18

Note: [Urdu instructions printed at top of paper — best-effort] ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ جواب میں ایک سے زیادہ دائرے بھرے ہوں تو جواب غلط تصور ہوگا۔

Q1. Select the star codon of the given.

(A) AUA
(B) AUG
(C) AUU
(D) AAA

Q2. Endosymbiosis hypothesis explain the origin of _____.

(A) Fish
(B) Bird
(C) Prokaryotes
(D) Eukaryotes

Q3. The rate at which the new individuals are added to a population in a unit time is _____.

(A) Density
(B) Natily
(C) Mortality
(D) Dispersion

Q4. Multiple cloning sites in a vector give flexibility in te choice of _____.

(A) Resistant gene
(B) Restriction enzymes
(C) Desired sequence
(D) Primers

Q5. Most common breed of buffalo in Pakistan is _____.

(A) Dajal
(B) Dhani
(C) Nili Ravi
(D) Sahiwal

Q6. The hormone that induce labour and controls labour via a positive feedback mechanism is _____.

(A) Oxytocin
(B) Estrogen
(C) ADH
(D) Progesterone

Q7. Dimorphin is the chemical name of _____.

(A) Alcohol
(B) Cannabis
(C) Nicotine
(D) Heroin

Q8. Positive feedback enable _____.

(A) Child birth
(B) Goiter formation
(C) Bone thickening
(D) Dwarfism

Q9. Thigmonastic movement occurs in response to _____.

(A) Gravity
(B) Touch
(C) Chemicals
(D) Light

Q10. The corpus leutium is formed at the site of _____.

(A) Fertilization
(B) Menstruation
(C) Ovulation
(D) Implanation

Q11. Genes located on the same chromosome and inheriting together are said to be _____.

(A) Crossed
(B) Dominant
(C) Recessive
(D) Linked

Q12. Genotype R_1R_2 of 4 O'clock plant produces _____ colour flowers.

(A) Red
(B) White
(C) Pink
(D) Yellow

Q13. Hyponutremia is _____.

(A) Low blood sodium
(B) Abscess
(C) Pus
(D) Purulent

Q14. The left lung of human is smaller in size because of asymmetrical position of _____.

(A) Liver
(B) Trachea
(C) Heart
(D) Intestines

Q15. Urine leave the body from bladder through a duct called _____.

(A) Ureter
(B) Urethra
(C) Vasarecta
(D) Hilus

Q16. How many unpaired bones are found in cranium?

(A) Two
(B) Three
(C) Four
(D) Eight

Q17. Which muscles are arranged in branching pattern?

(A) Skeletal muscles
(B) Smooth muscles
(C) Cardiac muscles
(D) Sphincter muscles

Q18. _____ bring nerve impulse toward soma of neuron.

(A) Axon
(B) Dendride
(C) Schwan cell
(D) Axon terminals

SECTION-B

Time Allowed: 2 Hours and 40 minutes

QQ2. Answer any TEN parts. Each part carries FOUR marks.

(1) State the symptoms and complicaions related to pneumonia?

(2) Describe about the components of Renal Tubules?

(3) Enumerate about the types and role of bone cells.

(4) How the developments of Active membrane potential occur during nerves impulse transmission?

(5) Why anterior pituitary is considered as master gland?

(6) Write any four differences between innate and learning behavior?

(7) Enlist the causes of miscarriage?

(8) Define lactation and write the advantages of breast feeding?

(9) What is ABO Blood Group System?

(10) Compare Lamarckism with Darwinism?

(11) Enlist the problems related to rapid growth in human population?

(12) Discuss the steps of Gel Electrophoresis in DNA sequencing?

(13) Give the importance of animal husbandry.

SECTION-C

Note: Attempt any THREE questions. All questions carry equal marks.

QQ3. Explain how Griffith and Avery proved that DNA is heredity material?

QQ4.

(a) Illustrate about the larynx and trachea?

(b) Describe the role of hypothalamus in thermoregulation?

QQ5.

(a) Note the differences between fibrous and synovial joints?

(b) Write a note on thyroid gland?

QQ6.

(a) Explain briefly about biological fixation of nitrogen?

(b) Write the functions of spinal cord?